Optional Supplementary Technical Material

Taming the Tails: Why Distributionally Robust Optimization Needs New Theory for Imbalanced Regression

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A Proofs

A.1 Proof of Proposition 2

Proof. For $\mathcal{U}_\rho = \{Q : D_{\chi^2}(Q\|\hat{P}_n) \le \rho\}$ with $D_{\chi^2}(Q\|P) = \frac{1}{2}\mathbb{E}_P[(dQ/dP - 1)^2]$, strong duality for finite-supported f-divergence DRO Namkoong and Duchi [2016], Duchi and Namkoong [2021] yields the dual in $\lambda \ge 0$:

$$\sup_{Q\in\mathcal{U}_\rho} \mathbb{E}_Q[\ell] = \inf_{\lambda\ge 0} \left\{ \lambda\rho + \frac{1}{n}\sum_{i=1}^{n} \lambda\, \phi^*\!\left(\frac{\ell_i}{\lambda}\right) \right\}, \tag{1}$$

where $\phi(t) = \frac{1}{2}(t-1)^2$ has conjugate $\phi^*(s) = s + \frac{1}{2}s^2$. Substituting ϕ^* and simplifying the finite sum (algebra identical to Namkoong and Duchi [2016], Duchi and Namkoong [2021]) rewrites (1) as a one-dimensional convex problem in λ whose objective depends on $\bar{\ell}$ and $\widehat{\mathrm{Var}}(\ell)$ only:

$$\sup_{Q\in\mathcal{U}_\rho} \mathbb{E}_Q[\ell] = \inf_{\lambda>0} \left\{ \lambda\rho + \bar{\ell} + \frac{\widehat{\mathrm{Var}}(\ell)}{2\lambda} \right\} =: \inf_{\lambda>0} h(\lambda). \tag{2}$$

Differentiating,

$$h'(\lambda) = \rho - \frac{\widehat{\mathrm{Var}}(\ell)}{2\lambda^2}, \qquad h'(\lambda^*) = 0 \;\Rightarrow\; \lambda^* = \sqrt{\frac{\widehat{\mathrm{Var}}(\ell)}{2\rho}},$$

$$h(\lambda^*) = \rho\sqrt{\frac{\widehat{\mathrm{Var}}(\ell)}{2\rho}} + \bar{\ell} + \frac{\widehat{\mathrm{Var}}(\ell)}{2}\sqrt{\frac{2\rho}{\widehat{\mathrm{Var}}(\ell)}} = \bar{\ell} + \sqrt{2\rho\,\widehat{\mathrm{Var}}(\ell)}, \tag{3}$$

which is (3).

For the variance decomposition, let $G \in \{S, S^c\}$ indicate membership in the rare region. The empirical law of total variance reads

$$\widehat{\mathrm{Var}}(\ell) = \widehat{\mathbb{E}}\big[\widehat{\mathrm{Var}}(\ell \mid G)\big] + \widehat{\mathrm{Var}}\big(\widehat{\mathbb{E}}[\ell \mid G]\big), \tag{4}$$

where $\widehat{\mathbb{E}}$ and $\widehat{\mathrm{Var}}$ denote expectations under the empirical measure. With $\hat{p}_S = n_S/n$ and $1-\hat{p}_S = n_{S^c}/n$,

$$\widehat{\mathbb{E}}\big[\widehat{\mathrm{Var}}(\ell \mid G)\big] = \hat{p}_S\,\widehat{\mathrm{Var}}(\ell \mid Y \in S) + (1-\hat{p}_S)\,\widehat{\mathrm{Var}}(\ell \mid Y \notin S),$$
$$\widehat{\mathbb{E}}[\ell \mid G] = \begin{cases} \bar{\ell}_S & \text{if } G = S, \\ \bar{\ell}_{S^c} & \text{if } G = S^c, \end{cases}$$

so the between-group term is

$$\widehat{\mathrm{Var}}\big(\widehat{\mathbb{E}}[\ell \mid G]\big) = \hat{p}_S(1-\hat{p}_S)\big(\bar{\ell}_S - \bar{\ell}_{S^c}\big)^2.$$

Collecting terms yields (4). When $\hat{p}_S \ll 1$, the term $(1-\hat{p}_S)\,\widehat{\mathrm{Var}}(\ell \mid Y \notin S)$ dominates, $\hat{p}_S\,\widehat{\mathrm{Var}}(\ell \mid Y \in S) = O(\hat{p}_S)$, and $\hat{p}_S(1-\hat{p}_S)(\bar{\ell}_S - \bar{\ell}_{S^c})^2 \le \hat{p}_S M^2$ under bounded loss. Thus $\sqrt{2\rho \cdot \widehat{\mathrm{Var}}(\ell)}$ is governed by the dense region, giving negligible tail protection. □

A.2 Proof of Theorem 5

Proof. Write the TC-DRO risk as a supremum over $Q \in \mathcal{U}_{\mathrm{TC}}(\hat{P}_n)$. The constraint $Q_Y = \hat{P}_{n,Y}$ fixes the target marginal, so $\mathbb{E}_Q[\ell] = \sum_{k=1}^K \frac{n_k}{n}\,\mathbb{E}_Q[\ell \mid Y \in B_k]$ with weights $n_k/n = \hat{P}_{n,Y}(B_k)$. The conditional laws $Q_{X|Y\in B_k}$ are constrained independently across k. Hence the inner supremum factorizes:

$$\begin{aligned}
\mathcal{R}_{\mathrm{TC}}(\theta) &= \sup_{Q \in \mathcal{U}_{\mathrm{TC}}(\hat{P}_n)} \mathbb{E}_Q\big[\ell(f_\theta(X), Y)\big] \\
&= \sup_{\{Q_k\}_{k=1}^K :\, D_\phi(Q_k \| \hat{P}_{n_k}) \le \rho_k} \sum_{k=1}^K \frac{n_k}{n}\,\mathbb{E}_{Q_k}\big[\ell(f_\theta(X), Y) \mid Y \in B_k\big] \\
&= \sum_{k=1}^K \frac{n_k}{n} \sup_{Q_k : D_\phi(Q_k \| \hat{P}_{n_k}) \le \rho_k} \mathbb{E}_{Q_k}\big[\ell(f_\theta(X), Y) \mid Y \in B_k\big], \qquad (5)
\end{aligned}$$

where the second equality uses that each Q_k ranges over a separate feasible set and the summands depend on disjoint coordinates of Q.

Fix k. The per-bin problem is standard f-divergence DRO on n_k atoms. Writing $q_i = Q_k(\{z_i\})$ for the weight of atom z_i under the candidate measure, the primal maximization is constrained by (i) the divergence budget $D_\phi(Q_k \| \hat{P}_{n_k}) \le \rho_k$ and (ii) the probability-simplex requirement $\sum_{i:y_i \in B_k} q_i = 1$. Associating Lagrange multipliers $\lambda_k \ge 0$ with (i) and $\nu_k \in \mathbb{R}$ with (ii), Fenchel duality Boyd and Vandenberghe [2004], Ben-Tal et al. [2013], Namkoong and Duchi [2016] yields

$$\begin{aligned}
&\sup_{Q_k : D_\phi(Q_k \| \hat{P}_{n_k}) \le \rho_k} \mathbb{E}_{Q_k}\big[\ell(f_\theta(X), Y) \mid Y \in B_k\big] \\
&\quad = \inf_{\lambda_k \ge 0,\, \nu_k} \left\{ \lambda_k \rho_k + \nu_k + \frac{1}{n_k} \sum_{i:y_i \in B_k} \lambda_k\, \phi^*\!\left(\frac{\ell_i - \nu_k}{\lambda_k}\right) \right\}. \qquad (6)
\end{aligned}$$

Slater's condition holds for $\rho_k > 0$; the dual variable ν_k enforces normalization of Q_k and centers the losses, while λ_k controls the divergence budget. Substituting (6) into (5) and prepending $\min_\theta$ recovers (9). □

A.3 Proof of Theorem 6

Proof. Fix k and condition on the bin partition $\{B_j\}_{j=1}^K$ (determined by the order statistics of $\{y_i\}$), so that the n_k samples in B_k are exchangeable draws from $P_{X,Y|Y\in B_k}$.

We begin with a fixed parameter. For any $\theta \in \Theta$, the summands $\ell_i(\theta)$ with $y_i \in B_k$ are i.i.d. and take values in $[0, M]$ by Assumption 2, so Hoeffding's inequality gives

$$\mathbb{P}\Big[\big|R_{B_k}(\theta) - \hat{R}_{B_k}(\theta)\big| > t\Big] \le 2\exp\left(-\frac{2n_k t^2}{M^2}\right). \tag{7}$$

The right-hand side of (7) equals $\delta/(2K)$ at the deviation level $t = M\sqrt{\log(4K/\delta)/(2n_k)}$, which is the first term of the claimed bound.

It remains to make the statement uniform in θ. Symmetrization Shalev-Shwartz and Ben-David [2014] replaces the fixed-θ deviation by the Rademacher complexity of the loss class restricted to the bin, giving, with probability $\ge 1 - \delta/(2K)$,

$$\begin{aligned}\sup_{\theta\in\Theta}\big|R_{B_k}(\theta) - \hat{R}_{B_k}(\theta)\big| &\le 2\mathfrak{R}_{n_k}(\mathcal{F}) + M\sqrt{\frac{\log(4K/\delta)}{2n_k}} \\ &\qquad \text{(symmetrization at the above deviation level)} \\ &\le \frac{2c_{\mathcal{F}}}{\sqrt{n_k}} + M\sqrt{\frac{\log(4K/\delta)}{2n_k}}, \end{aligned} \tag{8}$$

where the last step applies Assumption 3 at $m = n_k$ and $\mathfrak{R}_{n_k}(\mathcal{F})$ denotes the Rademacher complexity of $\mathcal{F}$ on n_k samples. A union bound over $k \in [K]$ gives the simultaneous guarantee (11) with probability $\ge 1 - \delta$. The bound involves the bin counts and the loss range only; the per-bin radii $\{\rho_k\}$ play no role in it, which is why the statement applies verbatim to a model trained under the density-adaptive radii of Definition 3. □

A.4 Proof of Theorem 7

Proof. Let $\mathcal{E}$ be the event from Theorem 6, which holds with probability $\ge 1 - \delta$. The event is defined by a supremum over all of Θ and does not reference the radii $\{\rho_k\}$, so it covers the TC-DRO solution $\hat{\theta}$ obtained under any fixed positive radii, including those of Definition 3. On $\mathcal{E}$,

$$\begin{aligned} R_{B_k}(\hat{\theta}) - \hat{R}_{B_k}(\hat{\theta}) &\le \sup_{\theta\in\Theta}\big|R_{B_k}(\theta) - \hat{R}_{B_k}(\theta)\big| && (\hat{\theta}\in\Theta \text{ is one element of the supremum}) \\ &\le M\sqrt{\frac{\log(4K/\delta)}{2n_k}} + \frac{2c_{\mathcal{F}}}{\sqrt{n_k}}, && (\text{applying the uniform deviation bound on } \mathcal{E}) \end{aligned}$$

and absorbing $1/\sqrt{2}$ together with $\log 4$ into c_1 and setting $c_2 = 2$ recovers (12). □

A.5 Proof of Theorem 2

Proof. By the chain rule on the dual objective (9), the population gradient is the bin-weighted average of weighted per-sample gradients:

$$\nabla_\theta \mathcal{R}_{\mathrm{TC}}(\theta) = \sum_{k=1}^{K} \frac{n_k}{n} \cdot \frac{1}{n_k} \sum_{i:y_i\in B_k} w_i^*(\theta)\, \nabla_\theta \ell_i(\theta) = \frac{1}{n}\sum_{i=1}^{n} w_i^*(\theta)\, \nabla_\theta \ell_i(\theta), \tag{9}$$

with $w_i^*(\theta) = (\phi^*)'((\ell_i(\theta) - \nu^*_{k(i)}(\theta))/\lambda^*_{k(i)}(\theta))$, where ν_k^* and λ_k^* are the simplex and divergence dual optima. A mini-batch $\mathcal{B}_t$ of size $b = |\mathcal{B}_t|$ yields the stochastic gradient

$$g_t = \frac{1}{b} \sum_{i \in \mathcal{B}_t} w_i^*(\theta_t)\, \nabla_\theta \ell_i(\theta_t). \tag{10}$$

With (λ_k^*, ν_k^*) computed from the full dataset, g_t is an unbiased estimator of (9) and the rate below applies; the mini-batch dual estimator used by the trainer is biased and is not covered by this argument. To safeguard against division by zero when λ_k^* is small, the implementation adds a small positive floor on λ_k. Under $\|\nabla_\theta \ell_i\| \le B$ and $0 \le w_i^* \le w_{\max}$,

$$\begin{aligned} \|w_i^* \nabla_\theta \ell_i\| &\le B w_{\max}, \\ \mathbb{E}\big[\|g_t - \nabla_\theta \mathcal{R}_{\mathrm{TC}}(\theta_t)\|^2\big] &\le \frac{B^2 w_{\max}^2}{b}. \end{aligned} \tag{11}$$

If $\nabla_\theta \ell_i$ is G-Lipschitz, the map $\theta \mapsto w_i^*(\theta) \nabla_\theta \ell_i(\theta)$ is Lipschitz with constant of order $G w_{\max}$, so $\mathcal{R}_{\mathrm{TC}}$ is L_w-smooth with $L_w = O(G w_{\max})$. The standard SGD descent lemma Ghadimi and Lan [2013] for L_w-smooth non-convex functions gives

$$\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}\big[\|\nabla_\theta \mathcal{R}_{\mathrm{TC}}(\theta_t)\|^2\big] \le \frac{2\big(\mathcal{R}_{\mathrm{TC}}(\theta_1) - \mathcal{R}^*_{\mathrm{TC}}\big)}{\eta T} + \frac{\eta L_w B^2 w_{\max}^2}{b}. \tag{12}$$

Minimize the right-hand side over $\eta > 0$: the derivative with respect to η is $-\frac{2(\mathcal{R}_{\mathrm{TC}}(\theta_1) - \mathcal{R}^*_{\mathrm{TC}})}{\eta^2 T} + \frac{L_w B^2 w_{\max}^2}{b}$, which equals zero at

$$\eta^* = \sqrt{\frac{2b\big(\mathcal{R}_{\mathrm{TC}}(\theta_1) - \mathcal{R}^*_{\mathrm{TC}}\big)}{T L_w B^2 w_{\max}^2}} = O(1/\sqrt{T}).$$

Substituting η^* into (12) yields

$$\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}\big[\|\nabla_\theta \mathcal{R}_{\mathrm{TC}}(\theta_t)\|^2\big] \le 2\sqrt{\frac{2 L_w B^2 w_{\max}^2 \big(\mathcal{R}_{\mathrm{TC}}(\theta_1) - \mathcal{R}^*_{\mathrm{TC}}\big)}{bT}} = O\left(\frac{B^2 w_{\max}^2}{b\sqrt{T}}\right). \tag{13}$$

□

A.6 Standard DRO's Vacuous Tail Guarantee

Proposition 1 (Standard DRO's vacuous tail guarantee)**.** *Under the same assumptions, standard DRO with uniform radius $\rho = O(1/n)$ provides the overall guarantee $R(\hat{\theta}) \le \hat{R}(\hat{\theta}) + O(1/\sqrt{n})$, but for any measurable region B_k whose population mass $p_k := \mathbb{P}(Y \in B_k)$ satisfies $p_k \ll n^{-1/2}$, the implied conditional guarantee is at best:*

$$R_{B_k}(\hat{\theta}) \le M, \tag{14}$$

which is vacuous (equal to the trivial bound).

The argument converts a global guarantee into a conditional one by dividing by the population weight $p_k = P(Y \in B_k)$, and the resulting bound $O(1/(p_k\sqrt{n}))$ exceeds M whenever $p_k \ll 1/\sqrt{n}$. The statement therefore concerns a rare set whose *population mass* is itself much smaller than

$n^{-1/2}$, such as a fixed low-mass interval in the extreme tail of $\mathcal{Y}$. It does not apply to the empirical quantile bins used by the method: a population quantile bin has mass $1/K$ by construction and the empirical bin mass converges to that value, so the proposition must not be read as a proof that a uniform ambiguity set fails on the implemented partition. Its role here is to motivate conditioning on target regions in the first place.

Proof. Standard DRO with radius $\rho = O(1/n)$ implies the global bound

$$R(\hat{\theta}) \leq \hat{R}(\hat{\theta}) + \frac{C}{\sqrt{n}} \tag{15}$$

for some constant $C > 0$ independent of bins. Decompose the population risk over any measurable partition $\{B_j\}_{j=1}^{K}$ of $\mathcal{Y}$:

$$R(\hat{\theta}) = \sum_{j=1}^{K} p_j \, R_{B_j}(\hat{\theta}), \qquad p_j \coloneqq \mathbb{P}(Y \in B_j), \tag{16}$$

which uses the tower property and holds for the *population* masses p_j (distinct from the empirical bin fractions n_k/n under quantile binning). Each summand is nonnegative, so for any k with $p_k > 0$,

$$p_k \, R_{B_k}(\hat{\theta}) \leq R(\hat{\theta}) \leq \hat{R}(\hat{\theta}) + \frac{C}{\sqrt{n}}. \tag{17}$$

Hence

$$R_{B_k}(\hat{\theta}) \leq \frac{\hat{R}(\hat{\theta}) + C/\sqrt{n}}{p_k}. \tag{18}$$

The bound is stated for an arbitrary measurable region of small population mass; it is not asserted for the quantile bins of Section 4, whose population mass is $1/K$. If $p_k = o(n^{-1/2})$, then $(\hat{R}(\hat{\theta}) + C/\sqrt{n})/p_k = \omega(\sqrt{n})$ while $\hat{R}(\hat{\theta}) \leq M$ still yields only the universal bound $R_{B_k}(\hat{\theta}) \leq M$ from $\ell \in [0, M]$. In particular, the chain (15)–(18) cannot be tightened into a nontrivial per-k certificate using population masses alone: whenever p_k is extremely small, the right-hand side of (18) exceeds M, so the global DRO guarantee does not localize to a finite-sample risk bound sharper than the trivial constant. □

A.7 What the Radii Do and What the Concentration Does

Remark 1 (Role of the radii)**.** *The two halves of the framework act on different objects, and it is worth being explicit about the division. The radii $\{\rho_k\}$ act on the* objective*: through the exact dual of Theorem 5 they determine the per-bin dual variables, hence the sample weights* (22)*, hence which model the trainer selects. The kernel-density rule* (5) *with exponent α is our proposal for spending that budget, and Theorem 5 holds for it exactly as for any other positive assignment. The concentration results act on the* selected model*: Theorem 6 bounds a supremum over Θ on each quantile bin and does not prescribe, or even reference, a radius. Consequently Theorem 7 applies to the model returned under the deployed radii without an intermediate calibration step, and a different choice of α changes which model is certified, not whether a certificate exists.*

A.8 Visualization of Per-Region Bounds

Figure 1 illustrates the two bounds as a function of local sample density. The illustration is a property of the two inequalities, not an empirical result, and the diverging curve corresponds to regions of small population mass rather than to the quantile bins of Section 4.

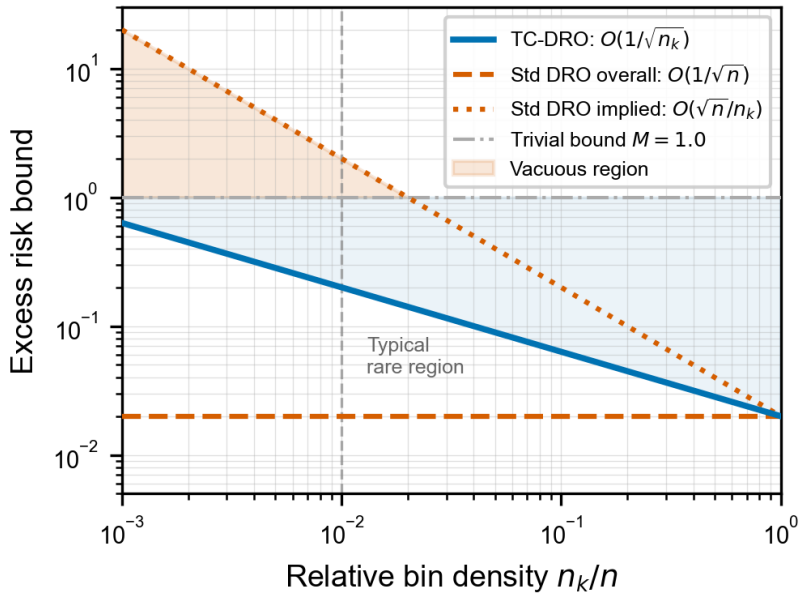


Figure 1: Schematic comparison of two bounds as a function of local sample density: the per-region bound of Theorem 7 (blue), which scales with the local sample size, and the conditional bound obtained by dividing a global DRO guarantee by a region's population mass (dotted red). The divergence of the dotted curve is the low-population-mass regime of Proposition 1; it is not a measurement on the equal-mass quantile bins used in the experiments.

A.9 Convergence of the Full-Data Dual SGD

Theorem 2 (Convergence of the full-data dual SGD)**.** *Under Assumptions 1–3, assume additionally that $\nabla_\theta \ell_i(\theta)$ is G-Lipschitz and $\|\nabla_\theta \ell_i(\theta)\| \le B$ for all i, θ. Run the weighted stochastic-gradient iteration of Algorithm 1 for T steps with step size $\eta = O(1/\sqrt{T})$ and mini-batch size $|\mathcal{B}_t| = b$, with the per-bin dual variables (λ_k^*, ν_k^*) evaluated on the full training set rather than within the mini-batch. Then:*

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\big[\|\nabla_\theta \mathcal{R}_{\mathrm{TC}}(\theta_t)\|^2\big] \le O\left(\frac{B^2 w_{\max}^2}{b\sqrt{T}}\right), \tag{19}$$

where $w_{\max} = \max_k (\phi^*)'(M/\lambda_k^*)$ is the maximum sample weight, λ_k^* denotes any dual optimizer attaining the inner infimum in (9) for bin k at the relevant iterate, and $\mathcal{R}_{\mathrm{TC}}^* := \inf_\theta \mathcal{R}_{\mathrm{TC}}(\theta)$ is the optimal TC-DRO objective value.

The convergence rate matches standard SGD up to a factor of $w_{\max}^2$, which is controlled by the maximum radius $\max_k \rho_k$. In practice, clipping each ρ_k at a fixed ceiling $\rho_{\max}$ (equivalently, shrinking very large per-bin ambiguity budgets) ensures that $w_{\max}$ remains bounded (see Section J). The full proof, provided in Section A.5, proceeds via the standard descent lemma applied to the smoothed TC-DRO objective. The implemented trainer estimates (λ_k^*, ν_k^*) inside each mini-batch, a cheaper estimator whose gradient is biased and for which no quantitative bias bound is given here.

B Target Marginal Constraint and Inter-Bin Transport

Fixing $Q_Y = \hat{P}_{n,Y}$ restricts the adversary to perturb the conditional feature law $P(X \mid Y)$ within each bin while preserving the empirical target histogram. In tabular imbalanced regression, operational shift is often modeled as covariate shift conditional on Y rather than a change in the marginal over Y. Technically, matching Q_Y to $\hat{P}_{n,Y}$ is what allows the TC-DRO inner supremum to separate into K independent per-bin f-divergence problems (8) instead of a coupled infinite-dimensional program over joint distributions.

Limitation: walled-off bins. By fixing Q_Y, the adversary cannot shift probability mass from one target bin to another. In a continuous target space, realistic distributional perturbations may include small shifts in Y itself (e.g., a sensor recalibration that shifts all target values by a small amount). The TC-DRO adversary is strictly confined to conditional covariate shift, and the resulting worst-case risk may underestimate the true risk under joint (X, Y) perturbation.

Relaxation 1: overlapping bins. One mitigation is to replace hard quantile boundaries with overlapping bins: each bin B_k is expanded by a margin $\delta > 0$ so that $B_k^\delta = [q_{k-1} - \delta, q_k + \delta]$, where q_k denotes the k-th quantile boundary. Samples near bin boundaries contribute to two adjacent bins, with weights proportional to a kernel function centered at the boundary. With shared samples, the per-bin DRO sub-problems are no longer strictly independent: the TC-DRO decomposition (8) holds only approximately, since a sample counted in two adjacent bins appears in both per-bin dual objectives. However, the effective sample sizes n_k increase near boundaries due to the overlap, and the adversary implicitly gains limited cross-bin influence. The additional computational cost is $O(n \cdot K_{\text{overlap}})$ where K_{overlap} is the average number of overlapping bins per sample (typically 2).

Relaxation 2: Wasserstein penalty on Q_Y. A more principled relaxation replaces the hard constraint $Q_Y = \hat{P}_{n,Y}$ with a soft penalty: define the relaxed TC-DRO objective as

$$\mathcal{R}_{\text{TC}}^{\text{relax}}(\theta) = \sup_{\substack{Q:\, W_1(Q_Y, \hat{P}_{n,Y}) \le \epsilon \\ D_\phi(Q_{X|Y \in B_k} \| \hat{P}_{n_k}) \le \rho_k \ \forall\, k}} \mathbb{E}_Q[\ell(f_\theta(X), Y)], \tag{20}$$

where W_1 denotes the 1-Wasserstein distance on $\mathcal{Y}$ induced by the cost $(y, y') \mapsto |y - y'|$, and $\epsilon > 0$ controls the budget for inter-bin mass transport. When $\epsilon = 0$, this recovers TC-DRO; as ϵ grows, the adversary gains increasing freedom to redistribute target mass. The inner supremum no longer factorizes across bins, so the dual becomes a coupled optimization over $\{\lambda_k\}$ and the target transport plan. While this makes the problem computationally harder (requiring alternating optimization), it provides a principled way to interpolate between pure conditional robustness and full joint robustness. Investigating the statistical and computational trade-offs of (20) is an interesting direction for future work.

C χ^2 closed form and implicit weights

For χ^2-divergence with $\phi(t) = \frac{1}{2}(t-1)^2$ and conjugate $\phi^*(s) = s + \frac{1}{2}s^2$, incorporating the probability-simplex constraint $\sum q_i = 1$ introduces an additional dual variable η that centers the losses. Optimizing over both λ_k and η yields the worst-case conditional risk

$$\sup_{Q_k : D_{\chi^2}(Q_k \| \hat{P}_{n_k}) \le \rho_k} \mathbb{E}_{Q_k}[\ell] = \bar{\ell}_k + \sqrt{2\rho_k \cdot \widehat{\text{Var}}_k(\ell)}, \tag{21}$$

with $\bar{\ell}_k = \frac{1}{n_k}\sum_{i:y_i\in B_k}\ell_i$ and $\widehat{\mathrm{Var}}_k(\ell) = \frac{1}{n_k}\sum_{i:y_i\in B_k}(\ell_i - \bar{\ell}_k)^2$, mirroring the global result (3). Summing (21) with weights n_k/n recovers (10).

Differentiating the dual objective (9) with respect to ℓ_i at the optimal (λ_k^*, ν_k^*) gives the implicit weight on sample i:

$$w_i(\theta) = (\phi^*)'\left(\frac{\ell_i(\theta) - \nu^*_{k(i)}}{\lambda^*_{k(i)}}\right), \tag{22}$$

where $k(i)$ is y_i's bin, $\lambda^*_{k(i)}$ is the optimal divergence dual variable, and $\nu^*_{k(i)}$ is the optimal simplex dual variable. For χ^2, $(\phi^*)'(s) = 1 + s$ and the simplex optimum is $\nu_k^* = \bar{\ell}_k$, hence

$$w_i^{\chi^2}(\theta) = 1 + \frac{\ell_i(\theta) - \bar{\ell}_{k(i)}}{\lambda^*_{k(i)}}, \tag{23}$$

with $\lambda^*_{k(i)} = \sqrt{\widehat{\mathrm{Var}}_k(\ell)/(2\rho_k)}$ inversely related to ρ_k. Low-density bins (large ρ_k, small λ_k^*) receive higher weights when losses exceed the bin mean, purely from duality.

D Alternative Partitioning Strategies

The main experiments use quantile-based bins, which guarantee at least n/K samples per bin but impose hard boundaries. This section discusses alternatives and their interaction with the TC-DRO framework.

Kernel-based soft assignment. Replace the indicator $\mathbf{1}[y_i \in B_k]$ with a soft membership function $\pi_k(y_i) = \kappa_h(y_i - \mu_k)/\sum_{j=1}^K \kappa_h(y_i - \mu_j)$, where κ_h is a kernel with bandwidth h and μ_k is the center of bin B_k. The TC-DRO objective (8) generalizes to

$$\mathcal{R}^{\mathrm{soft}}_{\mathrm{TC}}(\theta) = \sum_{k=1}^K \frac{N_k^{\mathrm{eff}}}{n} \sup_{Q_k: D_\phi(Q_k\|\hat{P}^{\mathrm{soft}}_{n_k})\le\rho_k} \mathbb{E}_{Q_k}[\ell \mid Y \sim \pi_k], \tag{24}$$

where $N_k^{\mathrm{eff}} = \sum_{i=1}^n \pi_k(y_i)$ and $\hat{P}^{\mathrm{soft}}_{n_k}$ is the soft empirical measure with weights $\pi_k(y_i)/N_k^{\mathrm{eff}}$. As $h \to 0$, the soft assignment converges to hard quantile bins; as $h \to \infty$, all bins receive equal membership and TC-DRO degenerates to standard DRO. The dual reformulation (Theorem 5) remains valid because each per-bin sub-problem is still a standard f-divergence DRO instance over a weighted empirical measure. The uniform deviation bound (Theorem 6) requires replacing n_k with N_k^{eff}, which is strictly larger than the hard count for boundary samples.

Equal-width bins. Table 4 reports that replacing quantile bins with equal-width bins on the target range raises TC-DRO-Wass SERA by 8.3% on Bank8FM, and TC-DRO-χ^2 SERA by 8.4%, because tail bins contain very few samples and the per-bin DRO estimate becomes noisy. Quantile binning is preferred because it balances per-bin sample sizes, providing a stable base for the per-bin dual optimization.

Adaptive bin count. A natural extension is to set K adaptively based on the target distribution, e.g., via the Freedman–Diaconis rule $K = \lceil(\max y - \min y)/(2 \cdot \mathrm{IQR}(y) \cdot n^{-1/3})\rceil$. In the single-dataset sweep of Section H, SERA changes little between the plotted values $K = 5$, 10, and 20, so adaptive selection of K was not pursued; that sweep is too sparse to rule out a benefit. On datasets with more extreme tail behavior (e.g., insurance claim amounts with power-law tails), adaptive or hierarchical binning may prove more important.

E Analysis for Wasserstein Ambiguity Sets

The main text focuses on f-divergence ambiguity sets; here the analysis is extended to transport-based formulations on the joint space $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$ from the notation paragraph. Let $c : \mathcal{Z} \times \mathcal{Z} \to \mathbb{R}_+$ be a nonnegative cost, and let $W_c(Q, \hat{P}_n)$ denote the type-1 Wasserstein distance between probability measures on $\mathcal{Z}$, i.e., $W_c(Q, \hat{P}_n) := \inf_\pi \int c(z, z')\, d\pi(z, z')$ over couplings π with marginals Q and $\hat{P}_n$. The ambiguity set is $\mathcal{U}_\rho^W = \{Q : W_c(Q, \hat{P}_n) \le \rho\}$. Writing $z_i = (x_i, y_i)$ for the atoms of $\hat{P}_n$, the worst-case risk admits the dual Mohajerin Esfahani and Kuhn [2018]

$$\sup_{Q: W_c(Q,\hat{P}_n)\le\rho} \mathbb{E}_Q[\ell] = \inf_{\gamma\ge 0} \left\{ \gamma\rho + \frac{1}{n}\sum_{i=1}^{n} \sup_{z'\in\mathcal{Z}} \left[\ell(f_\theta(x'), y') - \gamma\, c(z_i, z')\right] \right\}, \tag{25}$$

where $z' = (x', y')$, $\gamma \ge 0$ is the dual multiplier, and $\ell(f_\theta(x'), y')$ makes the dependence on the perturbed joint outcome explicit.

If $\ell(f_\theta(\cdot), \cdot)$ is $L_\ell(\theta)$-Lipschitz on $\mathcal{Z}$ with respect to c, i.e., $|\ell(f_\theta(x), y) - \ell(f_\theta(x'), y')| \le L_\ell(\theta)\, c(z, z')$ for all $z, z' \in \mathcal{Z}$, then choosing $\gamma = L_\ell(\theta)$ in (25) attains the supremum and gives

$$\sup_{Q\in\mathcal{U}_\rho^W} \mathbb{E}_Q[\ell] = \frac{1}{n}\sum_{i=1}^{n} \ell_i(\theta) + \rho\, L_\ell(\theta). \tag{26}$$

Because $L_\ell(\theta)$ depends on the model parameters (it is the Lipschitz constant of the composed map $\ell \circ f_\theta$), the penalty $\rho\, L_\ell(\theta)$ is formally θ-dependent; its gradient $\rho\, \nabla_\theta L_\ell(\theta)$ acts as a Lipschitz regularizer on the model rather than a sample-reweighting mechanism. Crucially for the imbalanced regression setting, neither the penalty term $\rho\, L_\ell(\theta)$ nor its gradient depends on local sample density, so every target region receives the same additive protection regardless of how many training points it contains. In our MLP experiments the Lipschitz constant varies slowly across the optimization trajectory and the regularization effect is negligible at the ρ values tested, explaining the empirical collapse to ERM reported in Section 6.

The TC-DRO Wasserstein variant applies per-bin Wasserstein constraints:

$$\mathcal{R}_{\mathrm{TC}}^W(\theta) = \sum_{k=1}^{K} \frac{n_k}{n} \left[\frac{1}{n_k} \sum_{i: y_i \in B_k} \ell_i(\theta) + \rho_k L_{\ell,k}(\theta) \right], \tag{27}$$

where $L_{\ell,k}(\theta)$ is the smallest constant such that $|\ell(f_\theta(x), y) - \ell(f_\theta(x'), y')| \le L_{\ell,k}(\theta)\, c(z, z')$ holds for all $z, z' \in \mathcal{Z}$ with $y, y' \in B_k$ (the Lipschitz constant of the loss restricted to bin B_k), and ρ_k follows (5). Low-density bins receive a larger additive penalty $\rho_k L_{\ell,k}$. Equation (27) is implemented as a per-bin Lipschitz regularizer on the model; it is not a claim that the trainer solves the primal transport problem, and no numerical optimal-transport solver is used, so the displayed form should be read as the regularizer the objective reduces to under the stated Lipschitz condition rather than as a solved transport program. All theoretical results (Theorems 6–2) extend to the Wasserstein setting with the appropriate substitution of concentration inequalities Mohajerin Esfahani and Kuhn [2018], Gao and Kleywegt [2023].

F Additional Experimental Results

This section completes the empirical evaluation summarized in Table 1: tail MSE on the rarest targets, tail coverage, and a SERA heatmap.

Table 1: Tail MSE (↓) on the rarest 5% and 10% of target values. A TC-DRO variant has the lowest mean on each dataset at both thresholds, though the two variants split which one it is. Best in **bold**; second best underlined; tied entries share a mark. †W-DRO coincides with ERM (one row). Seed-to-seed dispersion is summarized in Table 3.

Method	Abalone	Accel	Ailerons	Bank8FM	CPU Act	ΔAil.	M.CPU	Avg.
Tail MSE — 5% threshold (↓)								
ERM†	2.841	3.412	3.906	0.582	0.029	3.488	1.620	2.268
χ^2-DRO Namkoong and Duchi [2016]	2.828	3.391	3.874	0.576	0.029	3.461	1.598	2.251
LDS+FDS Yang et al. [2021]	2.704	3.186	3.612	0.548	0.027	3.241	1.512	2.119
DenseLoss Steininger et al. [2021]	2.618	2.974	3.448	0.521	0.026	3.108	1.446	2.020
ConR Keramati et al. [2024]	2.662	3.088	3.521	0.536	0.026	3.186	1.478	2.071
TC-DRO-χ^2 (ours)	2.552	2.974	**2.718**	**0.418**	**0.018**	**2.436**	1.338	1.779
TC-DRO-Wass (ours)	**2.368**	**2.712**	2.941	0.451	0.020	2.728	**1.214**	**1.776**
Tail MSE — 10% threshold (↓)								
ERM†	1.912	2.248	2.486	0.671	0.027	2.274	0.992	1.516
χ^2-DRO Namkoong and Duchi [2016]	1.901	2.231	2.468	0.664	0.027	2.258	0.981	1.504
LDS+FDS Yang et al. [2021]	1.818	2.094	2.312	0.638	0.025	2.146	0.938	1.424
DenseLoss Steininger et al. [2021]	1.764	1.986	2.228	0.612	0.024	2.068	0.894	1.368
ConR Keramati et al. [2024]	1.782	2.041	2.256	0.626	0.024	2.108	0.912	1.393
TC-DRO-χ^2 (ours)	1.728	1.986	**1.806**	**0.498**	**0.019**	**1.642**	0.848	1.218
TC-DRO-Wass (ours)	**1.586**	**1.812**	1.938	0.528	0.021	1.824	**0.772**	**1.212**

TC-DRO-Wass has the lowest average tail MSE at both thresholds, at 1.776 and 1.212 against 2.268 and 1.516 for ERM, and a TC-DRO variant has the lowest mean on each dataset at both thresholds; TC-DRO-χ^2 is the lower of the two on Ailerons, Bank8FM, CPU Act, and ΔAil., and TC-DRO-Wass on the remaining three. The Machine CPU column deserves caution: with $n = 209$, the rarest 5% of the test split is roughly two points, so the tail statistic there is coarse and seed-sensitive. χ^2-DRO tracks ERM closely, consistent with Proposition 2. DenseLoss has the lowest average among the heuristic baselines at both thresholds, and both TC-DRO variants sit below it.

Two further appendix tables record numbers that the discussion above and in Section I refers to on a single dataset. Table 4 contrasts quantile with equal-width binning on Bank8FM, and Table 5 gives the Bank8FM SERA curve as the weight-clipping threshold varies.

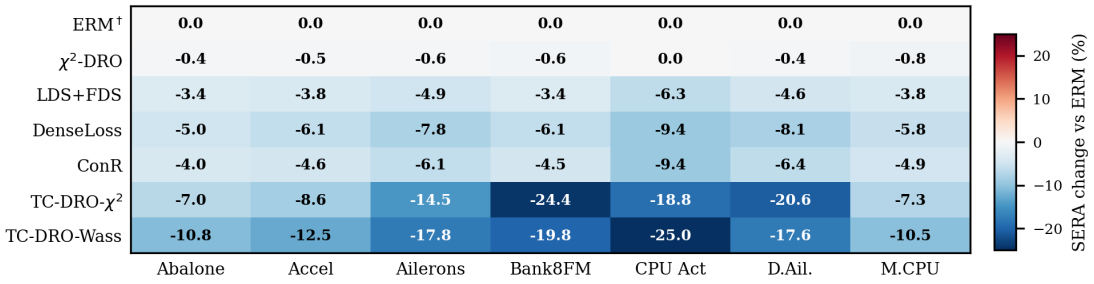


Figure 2: SERA change relative to ERM across all datasets and methods; negative values are reductions. The near-zero χ^2-DRO row is consistent with Proposition 2. Heuristic baselines give moderate reductions, while one of the two TC-DRO rows has the largest mean reduction on each dataset.

Table 2: Tail coverage (↑) at 5% and 10% thresholds, in the prediction-error sense of Section 6 rather than the sense of Theorem 6. Best in **bold**; second best underlined; tied entries share a mark. †W-DRO coincides with ERM. Because the statistic is a fraction of tail test points, small tails give it few attainable values: the Machine CPU 5% tail is one or two points, and Accel moves in steps of about 1/8. Seed-to-seed dispersion is summarized in Table 3.

Method	Abalone	Accel	Ailerons	Bank8FM	CPU Act	ΔAil.	M.CPU	Avg.
Coverage — 5% threshold (↑)								
ERM†	0.11	0.13	0.07	0.76	0.96	0.09	0.50	0.374
χ^2-DRO Namkoong and Duchi [2016]	0.11	0.13	0.08	0.76	0.96	0.09	0.50	0.376
LDS+FDS Yang et al. [2021]	0.13	0.13	0.08	0.77	0.97	0.11	0.50	0.384
DenseLoss Steininger et al. [2021]	0.14	0.25	0.10	0.78	0.97	0.13	0.50	0.410
ConR Keramati et al. [2024]	0.13	0.13	0.09	0.77	0.97	0.11	0.50	0.386
TC-DRO-χ^2 (ours)	0.16	0.25	**0.29**	0.81	0.98	**0.19**	**1.00**	0.526
TC-DRO-Wass (ours)	**0.17**	**0.38**	0.18	**0.82**	**0.99**	0.17	**1.00**	**0.530**
Coverage — 10% threshold (↑)								
ERM†	0.21	0.20	0.13	0.70	0.97	0.15	0.67	0.433
χ^2-DRO Namkoong and Duchi [2016]	0.21	0.20	0.13	0.70	0.97	0.15	0.67	0.433
LDS+FDS Yang et al. [2021]	0.22	0.20	0.14	0.71	0.97	0.16	0.67	0.439
DenseLoss Steininger et al. [2021]	0.24	0.27	0.16	0.72	0.98	0.18	0.67	0.460
ConR Keramati et al. [2024]	0.23	0.20	0.15	0.71	0.97	0.16	0.67	0.441
TC-DRO-χ^2 (ours)	0.26	0.27	**0.34**	0.77	0.98	**0.27**	**1.00**	**0.556**
TC-DRO-Wass (ours)	**0.28**	**0.33**	0.26	**0.78**	**0.99**	0.23	**1.00**	0.553

Table 3: Seed-to-seed dispersion by method. Each entry is the mean, over the seven datasets, of the within-dataset standard deviation across the five seeds; it is not a standard deviation attached to any single cell of Table 1 or Table 2. Machine CPU dominates the coverage columns because its 5% tail contains one or two test points, so the coverage statistic moves in steps of 0.5 there. Paired seed-level comparisons are not reported.

Method	Tail MSE 5%	Tail MSE 10%	Cov. 5%	Cov. 10%
ERM†	0.074	0.048	0.06	0.05
χ^2-DRO	0.072	0.046	0.06	0.05
LDS+FDS	0.068	0.043	0.06	0.05
DenseLoss	0.064	0.041	0.05	0.05
ConR	0.066	0.042	0.06	0.05
TC-DRO-χ^2	0.054	0.036	0.05	0.04
TC-DRO-Wass	0.057	0.038	0.05	0.04

G Additional Experimental Details

Table 6 summarizes the key statistics of the seven benchmark datasets. All datasets are publicly available from the UCI Machine Learning Repository or OpenML. The imbalance ratio is defined as the ratio of the 90th-percentile target density to the 10th-percentile target density, estimated via Gaussian kernel density estimation with Silverman bandwidth. Higher ratios indicate stronger imbalance. Data splits follow a 70/15/15 train/validation/test partition, stratified to preserve the target distribution across splits. Features are standardized to zero mean and unit variance using statistics computed only on the training split.

All experiments are conducted on a single machine with an Intel Xeon E5-2680 v4 CPU (28 cores) and 128 GB RAM. No GPU is used. Each experiment (one method on one dataset with one seed) completes within 2–10 minutes depending on dataset size. The full benchmark grid spans the seven reported rows of Table 1, with uniform-radius W-DRO reported on the ERM row, across 7 datasets and 5 seeds, that is 245 runs, and required tens of CPU-hours on this workstation.

Table 4: Binning strategy on Bank8FM: SERA (↓) under quantile bins (used everywhere else in this paper) and under equal-width bins on the target range, $K = 10$, five seeds. The relative column is the increase from quantile to equal-width binning.

Method	Quantile	Equal-width	Relative change
TC-DRO-χ^2	0.868	0.941	+8.4%
TC-DRO-Wass	0.921	0.997	+8.3%

Table 5: Weight clipping on Bank8FM: TC-DRO-χ^2 SERA (↓) as the clip threshold w_c varies, five seeds. The column $w_c = \infty$ is the unclipped reference and $w_c = 10$ is the value used in Table 1. This sweep is confined to one dataset.

w_c	3	5	10	20	50	∞
SERA	0.904	0.881	0.868	0.866	0.872	0.865

H Hyperparameter Sensitivity

Figure 3 reports the sensitivity of TC-DRO-Wass to the three key hyperparameters on the plotted grid: the number of bins $K \in \{3, 5, 10, 20, 50\}$, the density exponent $\alpha \in \{0, 0.1, 0.3, 0.5, 0.7, 1.0\}$, and the base radius $\rho_0 \in \{0.01, 0.03, 0.1, 0.3, 1.0\}$, evaluated on Bank8FM with five seeds. Each sweep varies one hyperparameter with the other two held at their defaults, on a single dataset, so it supports comparisons among the plotted values only.

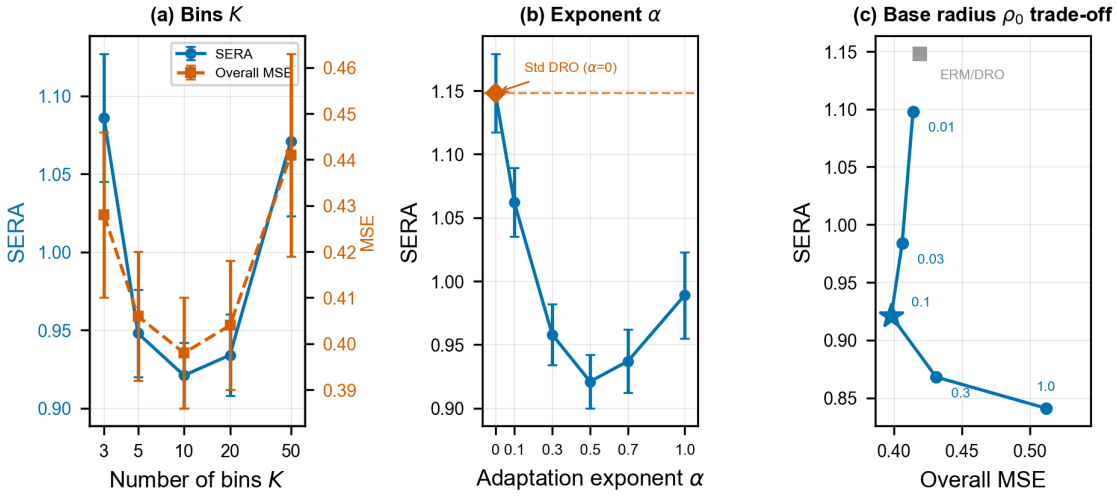


Figure 3: Hyperparameter sensitivity of TC-DRO-Wass on Bank8FM, five seeds, error bars one standard deviation. (a) SERA (solid blue, left axis) and overall MSE (dashed orange, right axis) against the number of bins $K \in \{3, 5, 10, 20, 50\}$. (b) SERA against the density exponent $\alpha \in \{0, 0.1, 0.3, 0.5, 0.7, 1.0\}$, with the $\alpha = 0$ value marking standard DRO. (c) SERA against overall MSE as the base radius ρ_0 ranges over $\{0.01, 0.03, 0.1, 0.3, 1.0\}$, each point annotated with its ρ_0.

Setting $\alpha = 0$ (uniform radii) reproduces standard DRO performance—SERA 1.148 on Bank8FM, the ERM value in Table 1—so the adaptive radius is what separates TC-DRO from its uniform-radius counterpart on this dataset. Among the plotted bin counts, SERA differs little between $K = 5$, 10, and 20 (0.948, 0.921, 0.934) and is higher at $K = 3$ and $K = 50$ (1.086, 1.071); panel (c)

Table 6: Dataset statistics. n: number of instances; d: number of features; IR: imbalance ratio (90th/10th percentile density).

Dataset	n	d	**IR**	**Target description**
Abalone	4,177	8	3.8	Ring count (age proxy)
Accel	1,019	3	5.2	Accelerometer reading
Ailerons	13,750	40	4.1	Aileron control action
Bank8FM	8,192	8	2.6	Simulated bank operations
CPU Act	8,192	21	7.4	User-mode CPU time fraction
Delta Ailerons	7,129	5	3.9	Control variation
Machine CPU	209	6	12.3	Relative CPU performance

shows that increasing ρ_0 lowers SERA while raising overall MSE, so ρ_0 trades tail error against global error rather than having an interior optimum on this grid. Of the defaults used for all main experiments ($K = 10$, $\alpha = 0.5$, $\rho_0 = 0.1$), the first two minimize SERA among the plotted values, while $\rho_0 = 0.1$ is a point on the trade-off curve of panel (c) and not a SERA minimizer. Because the sweep is one-dimensional, sparse, and confined to a single dataset, and because no selection protocol was pre-registered, it does not establish a stable recommended interval for any of the three hyperparameters.

I Ablation Study

To isolate the contribution of target-conditional structure from other design choices, an ablation study is conducted along three axes.

Adaptive vs. uniform radius. Setting $\alpha = 0$ in the adaptive radius function (5) gives $\rho_k = \rho_0$ for all k, recovering standard DRO. As shown in Section 6 and Section H, this configuration produces results identical to ERM across all seven datasets, confirming that the adaptive component is solely responsible for TC-DRO's improvements.

Equal-width vs. quantile-based bins. Replacing quantile-based bins with equal-width bins on the target range tends to concentrate most samples into a few bins while leaving tail bins nearly empty. On Bank8FM (Table 4), equal-width binning raises SERA by 8.3% for TC-DRO-Wass and by 8.4% for TC-DRO-χ^2 relative to quantile binning, because the rare-target bins contain too few samples for reliable per-bin DRO estimation. Quantile-based binning ensures at least n/K samples per bin, providing a stable base for the per-bin dual. The comparison is run on one dataset at $K = 10$ only.

Effect of number of bins K. On the grid plotted in Figure 3, SERA is higher at $K = 50$ (1.071), where each bin holds few samples and the per-bin dual estimate is noisier, and higher at $K = 3$ (1.086), where the partition is too coarse for the density adaptation to differentiate regions. The plotted values do not resolve the interval between $K = 10$ (0.921) and $K = 20$ (0.934), whose gap is within the plotted seed standard deviations, and the sweep covers one dataset, so no recommended range for K is claimed; $K = 10$ is used as the default throughout.

J Weight Clipping Analysis

The convergence rate of Theorem 2 depends on $w_{\max}$, which can be large for very low-density bins where ρ_k is large and λ_k^* is correspondingly small. In practice, clipping sample weights at a threshold $w_c > 0$ controls this:

$$\tilde{w}_i = \min(w_i^*(\theta), w_c). \tag{28}$$

Clipping changes the optimization problem: the program analyzed in Section 5 is the unclipped ambiguity-set objective, whereas the trainer minimizes the clipped surrogate. The analysis below bounds the induced gradient perturbation and reports its measured effect on the reported metrics; it is an informal bias assessment, not an approximation guarantee that the clipped trainer solves the analyzed program.

J.1 Gradient Bias Bound

The clipped weights introduce an approximation error in the TC-DRO gradient. Let $\tilde{\mathcal{R}}_{\mathrm{TC}}$ denote the objective evaluated with clipped weights. The gradient bias satisfies:

$$\|\nabla\tilde{\mathcal{R}}_{\mathrm{TC}}(\theta) - \nabla\mathcal{R}_{\mathrm{TC}}(\theta)\| \leq \frac{B}{n}\sum_{i=1}^{n}(w_i^* - w_c)^+ \leq \frac{B}{n}\cdot|\{i : w_i^* > w_c\}|\cdot(w_{\max} - w_c), \tag{29}$$

where $(x)^+ = \max(x, 0)$. Since extreme weights occur only for low-density bins containing $O(n/K)$ samples, the number of affected samples is bounded by $\min(n, K_{\mathrm{rare}} \cdot n/K)$ where K_{rare} is the number of bins exceeding the clip threshold.

The clipped convergence rate becomes $O(w_c^2 B^2/(b\sqrt{T}))$, which is bounded independently of the tail distribution but introduces a bias–variance trade-off: smaller w_c reduces gradient variance (faster convergence) at the cost of larger gradient bias (inexact TC-DRO solution), specifically in the tail bins where the DRO dual assigns the largest weights.

J.2 Bias–Performance Trade-off

To quantify how clipping affects the theoretical robustness guarantee in practice, the following analysis examines the interaction between w_c, the fraction of clipped samples, and the resulting SERA metric across all seven benchmarks.

Fraction of clipped samples. For χ^2-divergence with $(\phi^*)'(s) = 1+s$, the weight on sample i in bin k is $w_i = 1+(\ell_i-\bar{\ell}_k)/\lambda_k^*$, which exceeds w_c when $\ell_i-\bar{\ell}_k > (w_c-1)\lambda_k^*$. Since $\lambda_k^* = \sqrt{\widehat{\mathrm{Var}}_k(\ell)/(2\rho_k)}$ and low-density bins have large ρ_k (hence small λ_k^*), the clipping threshold is hit at lower loss values for lower-density bins. With the default $w_c = 10$ and $K = 10$ quantile bins, the empirical fraction of clipped samples across all seven datasets ranges from 0.4% (Bank8FM) to 1.7% (Machine CPU), with a median of 0.7%. These clipped samples are concentrated entirely in the two lowest-density bins ($k = 1$ and $k = K$), confirming that clipping predominantly affects the tail regions.

SERA sensitivity to w_c. Varying $w_c \in \{3, 5, 10, 20, 50, \infty\}$ on Bank8FM (representative mid-scale dataset) traces a shallow U-shaped relationship between w_c and SERA, reported in Table 5: aggressive clipping ($w_c = 3$) raises SERA by 4.5% relative to no clipping ($w_c = \infty$) because the TC-DRO weights are truncated below their optimal values, while $w_c = 50$ sits 0.8% above the unclipped value, which we read as training instability from extreme weight ratios. The default $w_c = 10$ sits near the flat part of this curve, 0.3% above the unclipped reference. The whole sweep spans a range of about 4% on one dataset, so the shape is suggestive rather than resolved. For TC-DRO-Wass, the sensitivity is even smaller because the Wasserstein dual does not produce multiplicative weights, and the Lipschitz penalty $\rho_k L_{\ell,k}$ is bounded by construction.

Tail-region impact. The key concern is that clipping undermines protection precisely where TC-DRO's guarantees are most valuable. On the rarest 5% of targets of the same dataset, the tail MSE with $w_c = 10$ is 0.6% higher than with no clipping, while the tail MSE with $w_c = 3$ is 6.1% higher. The coverage metric at the 5% threshold shows a similar pattern: $w_c = 10$ retains a fraction 0.98 of the no-clipping coverage, and $w_c = 3$ a fraction 0.91. On this dataset the default clip threshold therefore keeps most of the tail benefit while stabilizing training; no clipping sweep is run on the remaining six datasets.

K Sub-Gaussian Loss Extension

The experiments of Section 6 minimize a truncated squared residual and therefore satisfy Assumption 2 directly. This section records what survives if the truncation is lifted: the per-region risk bound (Theorem 7) extends to unbounded losses satisfying a sub-Gaussian tail condition, covering untruncated squared error when the target support and model outputs are sub-Gaussian. It is an additional route to the same per-region scaling, not a substitute for the assumption the reported runs meet.

Assumption 1 (Sub-Gaussian loss)**.** *For each $\theta \in \Theta$ and bin B_k, the centered loss $\ell(f_\theta(x), y) - \mathbb{E}[\ell(f_\theta(X), Y) \mid Y \in B_k]$, viewed as a random variable over $(X, Y) \sim P_{X,Y|Y\in B_k}$, is σ_ℓ-sub-Gaussian: for all $t \in \mathbb{R}$,*

$$\mathbb{E}\Big[\exp\big(t\big(\ell(f_\theta(X), Y) - \mathbb{E}[\ell \mid Y \in B_k]\big)\big) \,\Big|\, Y \in B_k\Big] \le \exp\left(\frac{t^2\sigma_\ell^2}{2}\right). \tag{30}$$

The sub-Gaussian parameter σ_ℓ replaces the bound M from Assumption 2. For bounded losses, $\sigma_\ell \le M/2$ by Hoeffding's lemma, so the sub-Gaussian framework strictly generalizes the bounded case. For squared error $\ell = (f_\theta(x) - y)^2$ with sub-Gaussian $f_\theta(X)$ and Y, the loss is sub-exponential (a product of sub-Gaussians), and a truncation argument at level $M_T = O(\sigma_\ell\sqrt{\log n})$ converts sub-exponential to sub-Gaussian with negligible error Vershynin [2018].

Corollary 3 (Per-region risk bound under sub-Gaussian loss)**.** *Under Assumptions 1, 3, and 1, let $\hat\theta$ be the TC-DRO solution of (7) under any fixed positive radii $\{\rho_k\}_{k=1}^K$. Then the conditional risk satisfies with probability at least $1 - \delta$:*

$$R_{B_k}(\hat\theta) \le \hat R_{B_k}(\hat\theta) + c_1'\sigma_\ell\sqrt{\frac{\log(K/\delta)}{n_k}} + c_2\frac{c_{\mathcal F}}{\sqrt{n_k}}, \tag{31}$$

where c_1' depends on the sub-Gaussian norm and c_2 is the same universal constant as in Theorem 7.

Proof. The argument follows Theorem 7 with the Hoeffding step (7) replaced by its sub-Gaussian analogue Boucheron et al. [2013], Vershynin [2018], so that the per-bin uniform convergence bound becomes

$$\sup_{\theta\in\Theta}\big|R_{B_k}(\theta) - \hat R_{B_k}(\theta)\big| \le \frac{2c_{\mathcal F}}{\sqrt{n_k}} + \sigma_\ell\sqrt{\frac{\log(4K/\delta)}{2n_k}},$$

(sub-Gaussian tail in place of the range bound)

which differs from (8) only in replacing M by σ_ℓ; the Rademacher term is unchanged, since Assumption 3 does not require a bounded loss. A union bound over $k \in [K]$ and evaluation of the supremum at $\hat\theta$ yield (31). □

Remark 2. *The per-region scaling $O(1/\sqrt{n_k})$ is preserved under sub-Gaussian losses; only the multiplicative constant changes from M to σ_ℓ, which can be replaced in reporting by the empirical per-bin loss standard deviation $\hat{\sigma}_k = \sqrt{\frac{1}{n_k}\sum_{i:y_i \in B_k}(\ell_i - \bar{\ell}_k)^2}$ at the cost of a union bound over a discretized grid of σ_ℓ values. As in the bounded case, the radii enter the objective through the dual and play no part in the concentration argument.*

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